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Heated Dynamic Seal Rotary Union for Delivery of Cryogenic Fluid

Abstract

A rotary union for use with a rotating platen that includes one or more lip seals is disclosed. The rotary union includes a rotary union shaft and a rotary union housing that surrounds the rotary union shaft. Heaters are disposed on the outer surfaces of the rotary union shaft and the rotary union housing. Additionally, a low thermal conductivity path is created between the center portion of the rotary union housing, where the lip seals are located, and the base of the rotary union housing, which contacts the cryogenic fluid. This low thermal conductivity path allows the lip seals to remain sufficiently warm so as to ensure good seal performance and reduced leakage.

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Background/Summary

FIELD

[0001] Embodiments of the present disclosure relate to rotary unions for delivery of cryogenic fluids to a rotating platen, and more particularly, to rotary unions including a heated lip seal.

BACKGROUND

[0002] In some workpiece processing systems, the workpiece is disposed on a platen, which can be rotated in multiple directions. This platen may also include electrical components, such as electrodes that are in communication with a power source. Additionally, this platen may be designed to heat or cool the workpiece. To achieve cooling, fluids travel toward the platen through fluid conduits. These fluid conduits are in communication with an external fluid source and an external fluid sink. Because the platen rotates, a rotary union is typically used to link the platen to the fluid source and sink.

[0003] The workpiece may be a semiconductor wafer, which is subjected to one or more processes while disposed on the platen. These processes may include etching, deposition and ion implantation.

[0004] In some particular embodiments, the platen is cooled to cold temperatures to enhance the process being performed on the workpiece. This may be achieved by passing a cold fluid through the fluid conduits in or near the platen. However, seals used in conventional rotary unions often fail to seal at cryogenic temperatures. For example, leakage may result from a shrinkage of the seal due to temperature changes, from the seal becoming brittle or less compliant due to the low temperatures.

[0005] Consequently, it would be beneficial if there was a rotary union that could withstand these extremely low temperatures without excessive leakage. Such a rotary union would make it possible to perform cryogenic processing of the workpiece using a rotating platen.

SUMMARY

[0006] A rotary union for use with a rotating platen that includes one or more heated lip seals is disclosed. The rotary union includes a rotary union shaft and a rotary union housing that surrounds the rotary union shaft. Heaters are disposed on the outer surfaces of the rotary union shaft and the rotary union housing. Additionally, a low thermal conductivity path is created between the center portion of the rotary union housing, where the lip seals are located, and the base of the rotary union housing, which contacts the cryogenic fluid. This low thermal conductivity path allows the lip seals to remain sufficiently warm so as to ensure good seal performance and reduced leakage.

[0007] According to one embodiment, a rotary union for carrying a cryogenic fluid is disclosed. The rotary union comprises a rotary union shaft surrounding an internal fluid channel; a rotary union housing surrounding the rotary union shaft having a base in communication with a fluid inlet; a housing heater disposed on the rotary union housing; and one or more lip seals disposed between an outer surface of the rotary union shaft and an inner surface of a center portion of the rotary union housing; wherein the center portion of the rotary union housing is attached to the base of the rotary union housing by a connecting portion, and wherein the connecting portion comprises a tortuous pathway so as to increase a length of a path from the base to the center portion. In some embodiments, a shaft heater is disposed on an outer surface of the rotary union shaft. In some embodiments, the connecting portion of the rotary union housing comprises a low thermal conductivity pathway having a thermal resistance of 10° K/Watt or greater. In some embodiments, the rotary union housing comprises an outward facing portion, the center portion, the connecting portion and the base; and wherein the housing heater is disposed on the outward facing portion of the rotary union housing. In some embodiments, a thermal conductivity of a pathway from the housing heater to the center portion is at least 5 times greater than a thermal conductivity of the tortuous pathway. In some embodiments, channels are formed in the rotary union shaft, such that the outer surface of the rotary union shaft where the one or more lip seals are disposed is warmer than a temperature of the inner surface of the rotary union shaft which contacts the internal fluid

channel. In certain embodiments, the outer surface of the rotary union shaft is at least 100° C. warmer than the temperature of the inner surface.

[0008] According to another embodiment, a rotary union for carrying a cryogenic fluid is disclosed. The rotary union comprises a rotary union shaft surrounding an internal fluid channel; a rotary union housing surrounding the rotary union shaft having a base in communication with a fluid inlet, the rotary union housing comprising an upper portion, a center portion, a connecting portion and a base; a housing heater disposed on an outer surface of the upper portion of the rotary union housing; and one or more lip seals disposed between the outer surface of the rotary union shaft and an inner surface of the center portion of the rotary union housing; wherein an inlet or an outlet for the cryogenic fluid is disposed in the base; and a ratio of a thermal conductivity of a path from the housing heater to the center portion of the rotary union housing to a thermal conductivity of a path from the base to the center portion of the rotary union housing is at least 5. In some embodiments, the ratio is at least 10. In some embodiments, the connecting portion of the rotary union housing comprises a tortuous pathway so as to increase a length of the path from the base to the center portion. In some embodiments, the outer surface of the rotary union shaft where the one or more lip seals are disposed is at least 100° C. warmer than a temperature of the base. In some embodiments, a shaft heater is disposed on an outer surface of the rotary union shaft. In some embodiments, channels are formed in the rotary union shaft, such that the outer surface of the rotary union shaft where the one or more lip seals are disposed is warmer than a temperature of the inner surface of the rotary union shaft which contacts the internal fluid channel.

[0009] According to another embodiment, a rotating platen assembly is disclosed. The rotating platen assembly comprises a platen base; a rotatable upper assembly rotatably coupled to the platen base about an axis of rotation, the rotatable upper assembly including a platen; and a first rotary union, comprising: a first rotary union shaft surrounding an internal fluid channel; a first rotary union housing surrounding the first rotary union shaft having a base in communication with a fluid inlet; a first housing heater disposed on the first rotary union housing; and one or more lip seals disposed between an outer surface of the first rotary union shaft and an inner surface of a center portion of the first rotary union housing; wherein the center portion of the first rotary union housing is attached to the base of the first rotary union housing by a connecting portion, and wherein the connecting portion comprises a tortuous pathway so as to increase a length of a path from the base to the center portion; wherein the first rotary union is disposed on a first side of the platen base such that the internal fluid channel is aligned with the axis of rotation, wherein the first rotary union is configured to receive cryogenic fluid through the fluid inlet and to deliver the cryogenic fluid to the platen or a thermal cooling plate disposed adjacent to the platen via a supply tube in communication with the internal fluid channel. In some embodiments, the rotating platen assembly comprises a second rotary union shaft surrounding an internal fluid channel; a second rotary union housing surrounding the second rotary union shaft having a base in communication with a fluid outlet; a second housing heater disposed on the second rotary union housing; and one or more lip seals disposed between an outer surface of the second rotary union shaft and an inner surface of a center portion of the second rotary union housing; wherein the center portion of the second rotary union housing is attached to the base of the second rotary union housing by a connecting portion, and wherein the connecting portion comprises a tortuous pathway so as to increase a length of a path from the base to the center portion; wherein the second rotary union is disposed on a second side of the platen base such that the internal fluid channel is aligned with the axis of rotation, wherein the second rotary union is configured to receive cryogenic fluid from the platen or the thermal cooling plate via a drain tube in communication with the internal fluid channel and to discharge the cryogenic fluid via the fluid outlet.

[0010] According to another embodiment, an ion implantation system is disclosed. The ion implantation system comprises a process chamber, housing the rotating platen assembly described above; an ion source to generate an ion beam; and one or more beamline components to direct the

ion beam from the ion source to the process chamber. In some embodiments, a first shaft heater is disposed on an outer surface of the first rotary union shaft and a second shaft heater is disposed on an outer surface of the second rotary union shaft. In some embodiments, channels are formed in the first rotary union shaft, such that an outer surface of the first rotary union shaft where the one or more lip seals are disposed is warmer than a temperature of the inner surface of the first rotary union shaft. In some embodiments, a ratio of a thermal conductivity of a path from the first housing heater to the center portion of the first rotary union housing to a thermal conductivity of a path from the base to the center portion of the first rotary union housing is at least 5. In some embodiments, the ratio is at least 10.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0011] For a better understanding of the present disclosure, reference is made to the accompanying drawings, which are incorporated herein by reference and in which:

[0012] FIG. 1 shows a ion implantation system that may utilize a cryogenic platen;

[0013] FIG. 2A shows a side view of the cryogenic platen;

[0014] FIG. 2B shows a rear view of the cryogenic platen;

[0015] FIG. 3 shows the cryogenic rotary union with heated lip seals used in the cryogenic platen according to one embodiment;

[0016] FIG. 4 shows an enlarged view of the rotary union and the components disposed therein; and

[0017] FIG. 5 shows the thermal profile of the components shown in FIG. 4.

DETAILED DESCRIPTION

[0018] As described above, in certain systems, it is desirable to have a workpiece processed at very low temperatures, while disposed on a rotating platen. Further, mechanical seals are prone to leakage due to the low temperatures.

[0019] FIG. 1 shows an ion implantation system that may be used with a cryogenic platen. An ion source **100** is used to generate an ion beam **150**. The ion source **100** may be an indirectly heated cathode (IHC) ion source. Alternatively, the ion source **100** may be a capacitively coupled plasma source, an inductively coupled plasma source, a Bernas source or another source. Thus, the type of ion source is not limited by this disclosure. Disposed outside and proximate the extraction aperture of the ion source **100** is the extraction optics **101**, which may comprise one or more electrodes.

[0020] Located downstream from the extraction optics **101** is a mass analyzer **110**. The mass analyzer **110** uses magnetic fields to guide the path of the extracted ion beam. The magnetic fields affect the flight path of ions according to their mass and charge. A mass resolving device **120** that has a resolving aperture **121** is disposed at the output, or distal end, of the mass analyzer **110**. By proper selection of the magnetic fields, only those ions in the ion beam **150** that have a selected mass and charge will be directed through the resolving aperture **121**. Other ions will strike the mass resolving device **120** or a wall of the mass analyzer **110** and will not travel any further in the system.

[0021] A collimator **130** may be disposed downstream from the mass resolving device **120**. The collimator **130** accepts the ions from the ion beam **150** that pass through the resolving aperture **121** and creates an ion beam formed of a plurality of parallel or nearly parallel beamlets. The output, or distal end, of the mass analyzer **110** and the input, or proximal end, of the collimator **130** may be a fixed distance apart. The mass resolving device **120** is disposed in the space between these two components.

[0022] Located downstream from the collimator **130** may be an acceleration/deceleration stage **140**. The acceleration/deceleration stage **140** is a beam-line lens component configured to

independently control deflection, deceleration, and focus of the ion beam. For example, the acceleration/deceleration stage **140** may be an electrostatic filter (EF). The ion beam **150** that exits the acceleration/deceleration stage **140** enters the process chamber **160**.

[0023] Thus, one or more beamline components are disposed between the ion source **100** and the process chamber **160** to direct and guide the ion beam from the ion source **100** to the process chamber **160**. These beamline components may include a mass analyzer, a mass resolving device, a collimator and/or an acceleration/deceleration stage.

[0024] A controller **180** may be in communication with one or more power supplies such that the voltage or current supplied by these power supplies may be monitored and/or modified. The controller **180** may include a processing unit, such as a microcontroller, a personal computer, a special purpose controller, or another suitable processing unit. The controller **180** may also include a non-transitory storage element, such as a semiconductor memory, a magnetic memory, or another suitable memory. This non-transitory storage element may contain instructions and other data that allows the controller **180** to perform the functions described herein.

[0025] A rotating platen assembly **200** is disposed within the process chamber **160**. FIG. 2A shows a side view of the rotating platen assembly **200**, while FIG. 2B shows a rear view. The rotating platen assembly **200** includes a platen base **220**. A rotatable upper assembly **230** is rotatably coupled to the platen base **220**. A motor **215** is disposed in the platen base **220** and includes a drive belt **217** that allows rotation of the rotatable upper assembly **230** about axis **240**.

[0026] Further, the rotatable upper assembly **230** includes a platen **210** to which the workpiece **216** is mounted. The rotatable upper assembly **230** may also include a housing **218** that contains other components. The platen **210** is rotatable about axis **250**, which is orthogonal to the surface of the platen **210**. In this figure, the rotatable upper assembly **230** also includes a thermal cooling plate **211** that is provided adjacent to the platen **210**. The thermal cooling plate **211** does not rotate about axis **250**. Rather, when the platen **210** rotates, the thermal cooling plate **211** may be linearly translated to create a small gap between the thermal cooling plate **211** and the platen **210** to reduce friction during rotation of the platen **210**. When the platen **210** has completed its rotation, the thermal cooling plate **211** is linearly translated to contact the platen **210**, allowing thermal conduction between the thermal cooling plate **211** and the platen **210**. The thermal cooling plate **211** may include thermal channels through which cryogenic fluid can flow within the thermal cooling plate **211**. The cryogenic fluid may be a gas or liquid. As cryogenic fluid flows through the thermal cooling plate **211**, a workpiece **216** disposed on the platen **210** may be cooled to a desired temperature. The thermal channels in the thermal cooling plate **211** may connect to one or more supply tubes **302** and drain tubes **303**, which, in turn, are each connected to a respective rotary union **300**, **305**. A rotary union **300** is connected to cryogenic fluid source through fluid inlet **301**, while the second rotary union **305** is connected to a cryogenic fluid sink through a fluid outlet **304**. In some embodiments, the cryogenic fluid may flow through a closed system, wherein the cryogenic fluid source and the cryogenic fluid sink are in communication with one another. For example, the cryogenic fluid source and cryogenic fluid sink may be a chiller, which receives the cryogenic fluid from fluid outlet **304**, cools the fluid and recirculates it through fluid inlet **301**. It will be appreciated that although thermal cooling plate **211** may be used, it is also contemplated that thermal cooling plate **211** may not be utilized. Thus, in some embodiments, the thermal channels may be integrated directly into the platen **210** such that the supply tubes **302** and the drain tubes **303** are in communication with the platen **210**.

[0027] As shown in FIG. 3, one end of the rotary union **300** is affixed to the platen base **220**. The rotary union **300** is in communication with a fluid inlet **301**, which is fixed in position, and a supply tube **302**, which rotates about axis **240**. The rotary union **300** includes a rotary union housing **310**, which is fixed in position and includes the connection for the fluid inlet **301**. The rotary union **300** may include a rotary union shaft **320**, which rotates and includes the connection for the supply tube **302**. Further, the rotary union shaft **320** also includes an internal fluid channel

322, through which the cryogenic fluid passes.

[0028] The rotary union shaft **320** may be rotated using any suitable means. For example, the rotating shaft assembly may be moved by motor **215**, such that movement of the rotatable upper assembly **230** causes rotation of the rotary union shaft **320**.

[0029] A shaft heater **321** is mounted on the rotary union shaft **320**. Electrical wires (not shown) are connected to the shaft heater **321**.

[0030] A housing heater **311** is mounted on the rotary union housing **310**.

[0031] Lastly, lip seals **350** are used to seal the gap between the rotary union housing **310** and the rotary union shaft **320**. A seal spacer **351** may be disposed in the gap to maintain the separation between these components.

[0032] FIG. **4** shows an enlarged view of the rotary union **300** and the components disposed therein. The rotary union shaft **320** forms part of the outer wall of the internal fluid channel **322**. Consequently, the inner surface of the rotary union shaft **320** is subjected to very low temperatures, such as less than -40° C. To generate the desired heat gradient between the inner surface of the rotary union shaft **320** and the outer surface of the rotary union shaft **320**, one or more channels **323** are disposed in the rotary union shaft **320**. One or more openings **324** are used to allow the one or more channels **323** to be in communication with the environment within the process chamber **160**. Note that the one or more channels **323** are separated from one another by thin partitions **325**. These thin partitions **325** serve to reduce heat transfer between the inner surface and the outer surface of the rotary union shaft **320**. Thus, the rotary union shaft **320** may be configured as a hollow cylinder, wherein one or more channels **323** are disposed in the cylinder. The rotary union shaft **320** rotates with the rotatable upper assembly **230** and the supply tube **302**.

[0033] The shaft heater **321** may be disposed on the outer surface of the rotary union shaft **320** near the exit of the internal fluid channel **322**. This location may be selected as it is distant from the internal fluid channel **322** and the base **312** of the rotary union housing **310**, which are at very low temperatures due to their contact with the cryogenic fluid.

[0034] The rotary union housing **310** comprises a base **312**, which is affixed to the platen base **220** and is in communication with the fluid inlet **301** to allow the flow of cryogenic fluid. The rotary union housing **310** also comprises an outward facing portion **313**. The outward facing portion **313** is that portion of the rotary union housing **310** that is closest to the exit of the internal fluid channel **322** and extends outward from the platen base **220**. Outward facing portion **313** may be shaped as a hollow cylinder. The housing heater **311** is affixed to the outer surface of the outward facing portion **313** of the rotary union housing **310**. Over at least a part of its length, the inner surface of the outward facing portion **313** may be concentric to the outer surface of the rotary union shaft **320**. Further, bearings **360** are disposed between the inner surface of the outward facing portion **313** and the outer surface of the rotary union shaft **320**. Bearing retaining flanges **361** may be disposed on either the inner surface of the outward facing portion **313** or the outer surface of the rotary union shaft **320** so as to retain the bearings **360** in position.

[0035] The outward facing portion **313** attaches to the center portion **314** of the rotary union housing **310**. Like the outward facing portion, the center portion **314** may be shaped as a hollow cylinder. Over at least a part of its length, the inner surface of the center portion **314** may be concentric to the outer surface of the rotary union shaft **320**. One or more lip seals **350** are disposed in the gap between the inner surface of the center portion **314** and the outer surface of the rotary union shaft **320**. The seal spacer **351** is also located in the gap between the inner surface of the center portion **314** and the outer surface of the rotary union shaft **320** and serves to maintain the desired spacing between the lip seals **350**. The seal spacer **351** may be PEEK (poly ether ether ketone) although other low friction plastics capable of operation at cryogenic temperatures may be used. Additionally, one or more seal retainers **352** may also be disposed in the gap between the inner surface of the center portion **314** and the outer surface of the rotary union shaft **320** to hold the lip seals **350** in position. Since the housing heater **311** is disposed on the outward facing portion

313 and the one or more lip seals **350** are located near the center portion **314**, there is a high thermal conductivity path between the outward facing portion **313** and the center portion **314**. [0036] The center portion **314** of the rotary union housing **310** attaches to the base **312** through a connecting portion **315** which forms a low thermal conductivity path, resulting from a small cross sectional area and long distance from the base **312** to the seal area. As used herein, the term “low thermal conductivity path” refers to a path having a thermal resistance that is 10° K/Watt or greater. In some embodiments, the thermal resistance may be 15° K/Watt or greater. In some embodiments, the thermal resistance may be 20° K/Watt or greater. In other embodiments, the thermal resistance may be 30° K/Watt or greater. Specifically, the connecting portion **315** is a low thermal conductivity pathway between the base **312**, which is at very cold temperatures, and the center portion **314**, which is in contact with the one or more lip seals **350**.

[0037] As is well known, the rate of heat transfer is directly proportional to the cross-section area of the pathway and inversely proportional to the length of that pathway. Thus, to achieve a low thermal conductivity pathway, the cross-sectional area of the connecting portion **315** may be reduced, as compared to the cross-sectional area of the center portion **314** and the outward facing portion **313**. In some embodiments, the cross-sectional area of the connecting portion **315** may be one half or less than one half of the cross-sectional area of the center portion **314**. In certain embodiments, the cross-sectional area of the connecting portion **315** may be $\frac{1}{4}$ of the cross-sectional area of the center portion **314** or less. Additionally, the length of the connecting portion **315** may be increased. This may be achieved by incorporating a tortuous pathway in the connecting portion **315**. This tortuous pathway may comprise one or more folds, which serves to extend the path between the base **312** and the center portion **314**. As shown in FIG. 4, in some embodiments, the connecting portion **315** may be a plurality of segments that extend in the axial direction, where the folds reverse the direction of the path. However, other embodiments are also possible. For example, the connecting portion **315** may have a plurality of segments that extend radially outward and inward, where the folds reverse the direction of the path. Further, the plurality of segments may extend diagonally, such that there is a radial and axial component. In certain embodiments, the thermal conductivity of the connecting portion **315** is at least 10 times less than the thermal conductivity of the outward facing portion **313** or the center portion **314**. In some embodiments, the ratio of the thermal conductivity of the path from the outward facing portion **313** to the center portion **314** to the thermal conductivity of the path from the base **312** to the center portion **314** may be 5 or greater. In certain embodiments, the ratio may be 10 or greater. In some example embodiments, the ratio may be between 10 and 20. Further, while some of the figures show two folds, it is understood that additional folds may be introduced.

[0038] FIG. 5 shows the effects of the different thermal conductivities. In this figure, the shaft heater **321** and the housing heater **311** are both enabled and serve to heat the rotary union shaft **320** and the outward facing portion **313** of the rotary union housing **310**, respectively. Additionally, cryogenic fluid is passing through the internal fluid channel **322**, which serves to cool the base **312** and the inner surface of the rotary union shaft **320**. The darkest shading is used to indicate the warmest components, while the lighter shading indicates colder components. The shading between these two extremes indicates an intermediate temperature.

[0039] Referring first to the rotary union shaft **320**, the inner surface of that shaft is cold, due to its contact with the internal fluid channel **322** carrying cryogenic fluid. However, the channels **323** and thin partitions **325** create a low thermal conductivity pathway in the radial direction between the inner surface and the outer surface. Additionally, the shaft heater **321** is disposed on the outer surface of the rotary union shaft **320**. Thus, a large thermal gradient is created in the radial direction in the rotary union shaft **320**. In some embodiments, this thermal gradient may be greater than 100° C., such as between 110° and 150° C. However, note as well, that the distance from the shaft heater **321** in the axial or horizontal direction also affects the temperature of the rotary union shaft **320**. Thus, the lower part of the rotary union shaft **320**, which is spatially distant from the shaft heater

321, may be at a temperature that is between the temperature at the inner surface and the temperature of the shaft heater **321**.

[0040] Turning next to the rotary union housing **310**, the base **312** is cold, such as between -175° C. and -155° C., due to its contact with the cryogenic fluid. Meanwhile, the outward facing portion **313** of the rotary union housing **310** is relatively warm, such as greater than -25° C., due to the presence of the housing heater **311**. The outward facing portion **313** and the center portion **314** have relatively thick cross-sectional areas, allowing high thermal conductivity between these two portions. Consequently, the housing heater **311** also serves to warm the center portion **314**.

[0041] Therefore, the area of the rotary union housing **310** that contacts the one or more lip seals **350** is warmed by the housing heater **311**, while the area of the rotary union shaft **320** that contacts the one or more lip seals **350** is warmed by shaft heater **321**. Additionally, the low thermal conductivity pathway in the connecting portion **315** serves to reduce the flow of heat from the center portion **314** to the base **312**, allowing the center portion **314** to remain at the desired temperature. Thus, the center portion **314** may be more than 100° C. warmer than the base **312**.

[0042] The system described herein has many advantages. This system provides a superior technique for dynamically sealing a cryogenic fluid from the environment for use in a rotary union. By creating thermal isolation of a lip seal from the cryogenic fluid by means of a low thermal conductivity path between the cryogenic fluid and the lip seal, the seal temperature is maintained at a sufficiently high temperature such that good seal performance is achieved. This minimizes shrinkage of the lip seals due to thermal contraction, and allows the lip seals to maintain contact with the rotary union housing and the rotary union shaft. This also results in a longer life between servicing and a much lower leak rate to the environment compared to other dynamic seal technologies.

[0043] Note that FIGS. **3-5** show the rotary union that receives the cryogenic fluid through the fluid inlet **301** and delivers it to the platen **210**. The second rotary union is configured in a same manner, but is the mirror image of that one shown in FIG. **3-5**. In other words, if FIG. **3-5** show the rear view of the rotary union **300** in FIG. **2B**, these would be front views of the second rotary union **305**. Further, in second rotary union **305**, the flow of cryogenic fluid is in the opposite direction. Cryogenic fluid flows into the internal fluid channel **322** from the drain tube **303** and exits through the fluid outlet **304** in the base **312** to a cryogenic fluid sink.

[0044] The present disclosure is not to be limited in scope by the specific embodiments described herein. Indeed, other various embodiments of and modifications to the present disclosure, in addition to those described herein, will be apparent to those of ordinary skill in the art from the foregoing description and accompanying drawings. Thus, such other embodiments and modifications are intended to fall within the scope of the present disclosure. Furthermore, although the present disclosure has been described herein in the context of a particular implementation in a particular environment for a particular purpose, those of ordinary skill in the art will recognize that its usefulness is not limited thereto and that the present disclosure may be beneficially implemented in any number of environments for any number of purposes. Accordingly, the claims set forth below should be construed in view of the full breadth and spirit of the present disclosure as described herein.

Claims

1. A rotary union for carrying a cryogenic fluid, comprising: a rotary union shaft surrounding an internal fluid channel; a rotary union housing surrounding the rotary union shaft having a base in communication with a fluid inlet; a housing heater disposed on the rotary union housing; and one or more lip seals disposed between an outer surface of the rotary union shaft and an inner surface of a center portion of the rotary union housing; wherein the center portion of the rotary union housing is attached to the base of the rotary union housing by a connecting portion, and wherein the

connecting portion comprises a tortuous pathway so as to increase a length of a path from the base to the center portion.

2. The rotary union of claim 1, further comprising a shaft heater disposed on an outer surface of the rotary union shaft.

3. The rotary union of claim 1, wherein the connecting portion of the rotary union housing comprises a low thermal conductivity pathway having a thermal resistance of 10° K/Watt or greater.

4. The rotary union of claim 1, wherein the rotary union housing comprises an outward facing portion, the center portion, the connecting portion and the base; and wherein the housing heater is disposed on the outward facing portion of the rotary union housing.

5. The rotary union of claim 1, wherein a thermal conductivity of a pathway from the housing heater to the center portion is at least 5 times greater than a thermal conductivity of the tortuous pathway.

6. The rotary union of claim 1, wherein channels are formed in the rotary union shaft, such that the outer surface of the rotary union shaft where the one or more lip seals are disposed is warmer than a temperature of the inner surface of the rotary union shaft which contacts the internal fluid channel.

7. The rotary union of claim 6, wherein the outer surface of the rotary union shaft is at least 100° C. warmer than the temperature of the inner surface.

8. A rotary union for carrying a cryogenic fluid, comprising: a rotary union shaft surrounding an internal fluid channel; a rotary union housing surrounding the rotary union shaft having a base in communication with a fluid inlet, the rotary union housing comprising an upper portion, a center portion, a connecting portion and a base; a housing heater disposed on an outer surface of the upper portion of the rotary union housing; and one or more lip seals disposed between the outer surface of the rotary union shaft and an inner surface of the center portion of the rotary union housing; wherein an inlet or an outlet for the cryogenic fluid is disposed in the base; and a ratio of a thermal conductivity of a path from the housing heater to the center portion of the rotary union housing to a thermal conductivity of a path from the base to the center portion of the rotary union housing is at least 5.

9. The rotary union of claim 8, wherein the ratio is at least 10.

10. The rotary union of claim 8, wherein the connecting portion of the rotary union housing comprises a tortuous pathway so as to increase a length of the path from the base to the center portion.

11. The rotary union of claim 10, wherein the outer surface of the rotary union shaft where the one or more lip seals are disposed is at least 100° C. warmer than a temperature of the base.

12. The rotary union of claim 8, further comprising a shaft heater disposed on an outer surface of the rotary union shaft.

13. The rotary union of claim 12, wherein channels are formed in the rotary union shaft, such that the outer surface of the rotary union shaft where the one or more lip seals are disposed is warmer than a temperature of the inner surface of the rotary union shaft.

14. A rotating platen assembly, comprising: a platen base; a rotatable upper assembly rotatably coupled to the platen base about an axis of rotation, the rotatable upper assembly including a platen; and a first rotary union, comprising: a first rotary union shaft surrounding an internal fluid channel; a first rotary union housing surrounding the first rotary union shaft having a base in communication with a fluid inlet; a first housing heater disposed on the first rotary union housing; and one or more lip seals disposed between an outer surface of the first rotary union shaft and an inner surface of a center portion of the first rotary union housing; wherein the center portion of the first rotary union housing is attached to the base of the first rotary union housing by a connecting portion, and wherein the connecting portion comprises a tortuous pathway so as to increase a length of a path from the base to the center portion; wherein the first rotary union is disposed on a first

side of the platen base such that the internal fluid channel is aligned with the axis of rotation, wherein the first rotary union is configured to receive cryogenic fluid through the fluid inlet and to deliver the cryogenic fluid to the platen or a thermal cooling plate disposed adjacent to the platen via a supply tube in communication with the internal fluid channel.

15. The rotating platen assembly of claim 14, further comprising: a second rotary union shaft surrounding an internal fluid channel; a second rotary union housing surrounding the second rotary union shaft having a base in communication with a fluid outlet; a second housing heater disposed on the second rotary union housing; and one or more lip seals disposed between an outer surface of the second rotary union shaft and an inner surface of a center portion of the second rotary union housing; wherein the center portion of the second rotary union housing is attached to the base of the second rotary union housing by a connecting portion, and wherein the connecting portion comprises a tortuous pathway so as to increase a length of a path from the base to the center portion; wherein the second rotary union is disposed on a second side of the platen base such that the internal fluid channel is aligned with the axis of rotation, wherein the second rotary union is configured to receive cryogenic fluid from the platen or the thermal cooling plate via a drain tube in communication with the internal fluid channel and to discharge the cryogenic fluid via the fluid outlet.

16. An ion implantation system comprising: a process chamber, housing the rotating platen assembly of claim 15; an ion source to generate an ion beam; and one or more beamline components to direct the ion beam from the ion source to the process chamber.

17. The ion implantation system of claim 16, further comprising a first shaft heater disposed on an outer surface of the first rotary union shaft and a second shaft heater disposed on an outer surface of the second rotary union shaft.

18. The ion implantation system of claim 16, wherein channels are formed in the first rotary union shaft, such that an outer surface of the first rotary union shaft where the one or more lip seals are disposed is warmer than a temperature of the inner surface of the first rotary union shaft.

19. The ion implantation system of claim 16, wherein a ratio of a thermal conductivity of a path from the first housing heater to the center portion of the first rotary union housing to a thermal conductivity of a path from the base to the center portion of the first rotary union housing is at least 5.

20. The ion implantation system of claim 19, wherein the ratio is at least 10.
